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Vehicle transmission and method of operation

Abstract

A vehicle transmission using a gearset. Multi-mode clutches selectively couple components of the gearset. The transmission operates in and shifts between a first forward gear and a second forward gear using a change in input rotation. An electric motor coupled to the input may be used to change the input rotation direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/940,217 filed on Sep. 8, 2022. This application claims the benefit of U.S. Provisional Application No. 63/249,279, filed Sep. 28, 2021. The disclosures of the above applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention
- (1) The invention generally relates to a vehicle power train or drive system; and, more specifically, a drive system using an electric motor.
2. Description of Related Art
- (2) Vehicle powertrains or drive systems typically incorporate multiple clutch elements. Existing powertrains are configured as concentric and parallel axis architectures, including electric motors.
- (3) In the field of automotive technology, motor vehicle drive trains using electric motors and planetary drive trains include controllable or selectable coupling assemblies, for example, one-way clutches. These coupling assemblies can be electromagnetically operated and magnetically controlled.
- (4) These one-way clutches often include first and second members and at least one locking element (e.g., a strut, a pawl, etc.). The locking element moves between a deployed position, wherein the locking element extends from the first member and engages the second member, and a non-deployed position, wherein the locking element does not extend from the first member. Thereby, the first and second members are disengaged from each other.
- (5) Examples of one-way clutches can be found in U.S. Pat. No. 5,927,455, a bi-directional overrunning pawl-type clutch; U.S. Pat. No. 6,244,965 a planar overrunning coupling; and U.S. Pat. No. 6,290,044, a selectable one-way clutch assembly for use in an automatic transmission. U.S. Pat. Nos. 7,258,214 and 7,344,010 disclose overrunning coupling assemblies and U.S. Pat. No. 7,484,605 discloses an overrunning radial coupling assembly or clutch. The disclosures of each are hereby incorporated by reference. The foregoing is not exclusive; other selective or one-way clutches may be used and are known. The foregoing are examples of one-way clutches that may be

used in the vehicle drive system disclosed herein.

SUMMARY OF THE INVENTION

(6) A vehicle drive system having an electric motor operable in a first direction and a second direction. The system includes a first shaft, a second shaft, a first gearset connecting the first shaft to the second shaft, the first gearset including an intermediate gear, and a second gearset connecting the first shaft to the second shaft. A coupling assembly that selectively couples the first shaft to the second shaft through one of the first gearset and the second gearset. Wherein operating the electric motor in the first direction and selectively coupling the first shaft to the second shaft through the first gearset operates the second shaft at a first forward speed and operating the electric motor in the second direction and selectively coupling the first shaft to the second shaft through the second gearset operates the second shaft at a second forward speed.

(7) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. The detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. 1 is a schematic illustration of a vehicle drive system for a motor vehicle.

(3) FIG. 2 is a schematic illustration of an example gearset of a multi-speed transmission for use with the vehicle drive system of FIG. 1.

(4) FIG. 2A is a lever diagram of the gearset shown in FIG. 2.

(5) FIG. 3A is a side view of a controllable mechanical diode.

(6) FIG. 3B is a cross-sectional perspective view of a controllable mechanical diode.

(7) FIG. 4A is a partial view of a controllable mechanical diode showing the struts in their respective pockets.

(8) FIG. 4B is a partial view of a controllable mechanical diode showing the struts extending from their respective pockets.

(9) FIG. 5 is a table showing strut position associated with multiple controllable mechanical diode conditions.

(10) FIG. 6 is a schematic cross-sectional illustration of a 2-speed transmission in the park condition.

(11) FIG. 6A is a lever diagram of the gearset of the transmission shown in FIG. 6.

(12) FIG. 7 is a schematic cross-sectional illustration of a 2-speed transmission in a neutral condition.

(13) FIG. 7A is a lever diagram of the gearset of the transmission shown in FIG. 7.

(14) FIG. 8 is a schematic cross-sectional illustration of a 2-speed transmission operating in a 1st forward gear with power on and regeneration allowed.

(15) FIG. 8A is a lever diagram of the gearset of the transmission shown in FIG. 8.

(16) FIG. 9 is a schematic cross-sectional illustration of a 2-speed transmission in a 1st reverse gear with power on and regeneration allowed.

(17) FIG. 9A is a lever diagram of the gearset of the transmission shown in FIG. 9.

(18) FIG. 10 is a schematic cross-sectional illustration of a 2-speed transmission shifting from the 1.sup.st forward gear to the 2.sup.nd forward gear.

(19) FIG. 10A is a lever diagram of the gearset of the transmission shown in FIG. 10.

(20) FIG. 11 is a schematic cross-sectional illustration of a 2-speed transmission and the first and

- second CMDs **26**, **28** during shifting from the 1.sup.st forward gear to the 2.sup.nd forward gear
- (21) FIG. **11A** is a lever diagram of the gearset of the transmission shown in FIG. **11**.
- (22) FIG. **12** is a schematic cross-sectional illustration of a 2-speed transmission shifted from the 1.sup.st forward gear to the 2.sup.nd forward gear and operating in the 2.sup.nd forward gear with power on and regeneration allowed.
- (23) FIG. **12A** is a lever diagram of the gearset of the transmission shown in FIG. **12**.
- (24) FIGS. **13A-13E** are lever diagrams of the gearset of the transmission.
- (25) FIG. **14** is a schematic illustration of an additional embodiment of a planetary gearset according to an aspect of the present invention.
- (26) FIG. **14A** is a lever diagram of the gearset of the transmission shown in FIG. **14**.
- (27) FIGS. **15A-F** are lever diagrams of the gearset of the transmission shown in FIG. **14**.
- (28) FIG. **16** is a schematic illustration of an alternative embodiment of a two-speed transmission according to the present invention.
- (29) FIG. **16A** is a schematic illustration of another alternative embodiment of a two-speed transmission according to the present invention.
- (30) FIG. **17** is a schematic illustration of an additional alternative embodiment of a two-speed transmission according to the present invention.
- (31) FIG. **17A** is a lever diagram of the gearset of the transmission shown in FIG. **17**.
- (32) FIG. **18** is a schematic illustration of a further alternative embodiment of a two-speed transmission according to the present invention.
- (33) FIG. **18A** is a lever diagram of the gearset of the transmission shown in FIG. **18**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (34) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or its uses.
- (35) FIG. **1** is a schematic illustration of an electric vehicle drive system, generally indicated at **8**. The vehicle drive system **8** includes an electric motor **10** with two rotary output directions. For example, the motor output shaft spins in two pre-determined directions: clockwise or counterclockwise.
- (36) A transmission **12** connects the electric motor **10** to the vehicle wheels **14** through a differential **16**. The electric motor **10** operates as a motor in a power mode—receiving electrical power from the battery **18** and transforming it to mechanical power. The motor output of the electric motor **10** in the power mode provides the input to the transmission **12**. The transmission **12** provides an output to the vehicle wheels **14**. A vehicle control unit **20** interfaces with the electric motor **10**, transmission **12**, and battery **18** and controls the functional aspects of the electric powertrain.
- (37) In the power mode, the electric motor **10** motor receives electrical power from a battery **18** and transforms it to mechanical power. The mechanical power is the rotary input, clockwise or counterclockwise, to the transmission **12**.
- (38) In a regeneration mode—the electric motor **10** operates as a generator—receiving mechanical power from the transmission **12** and transforming it to electrical power. When an electric vehicle operates in a regeneration mode, the electric motor **10** feeds current to the battery **18** while developing a counter torque that acts as a brake. Said another way, on deceleration, manual rotation of the vehicle wheels **14** turns the electric motor **10**, essentially changing it into a generator, which creates electricity to charge the battery.
- (39) FIG. **2** schematically illustrates one example of the transmission **12**, as a 2-speed transmission including two simple planetary gearsets **22**, **24**, and two controllable mechanical diodes (CMD) **26**, **28**. As illustrated in FIG. **2**, the controllable mechanical diodes **26**, **28** located on each side of the planetary gearsets **22**, **24** operate to selectively couple portions of the planetary gearsets **22**, **24** to a transmission housing, generally indicated at **30**.
- (40) Selectively coupling portions of the planetary gearsets **22**, **24** to the transmission housing **30**

controls the transmission **12** output. The vehicle control unit **20** directly controls the controllable mechanical diodes (CMD) **26, 28** wherein each controllable mechanical diode (CMD) **26, 28** changes between multiple modes in response to a control signal from the vehicle control unit **20**.

(41) A controllable mechanical diode (CMD) may include a selectable one-way clutch. A selectable one-way clutch is a clutch that allows the selection of different modes. In one example, at least one clutch mode must be a one-way clutch. The clutches can either be static (1 race of the clutch not able to rotate) or dynamic (both clutch races can rotate). The one-way clutch modes allow for a smooth shift due to the synchronization characteristics of one-way clutches. While CMDs are disclosed, other selectable clutches, for example, a dynamically controllable clutch (DCC), along with other actuator travel clutches may also be used.

(42) FIGS. **3a-3b** schematically illustrate one example of the controllable mechanical diodes (CMD) **26, 28**. Each controllable mechanical diode (CMD) is a pair of one-way clutches (OWC) **32, 34** located on opposite sides of an annular plate **36** having internal splines **38**. One-way clutch **32** includes struts **40** operable to engage the annular plate **36** in one direction of rotation. One-way clutch **34** includes struts **42** operable to engage the annular plate **36** in the opposite direction of rotation.

(43) FIGS. **4a-4b** show the one-way clutch **32**, including a plurality of outwardly pivotable struts **40** mounted in recesses or pockets **46** in a side surface **48** of a housing **44** of the one-way clutch **32**. FIG. **4a** shows a selector plate **50**, located adjacent to the side surface **48** of the housing **44**, that retains, or keeps, the struts **40** in the recesses or pockets **46** in the housing **44**. An actuator **52**, see FIGS. **3a-3b**, coupled to the selector plate **50** by an arm **54**, moves or rotates the selector plate **50** in a counterclockwise direction, arrow **64**, to align the respective openings **58** in the selector plate **50** with the corresponding struts **40**. When the openings **58** in the selector plate **50** and the corresponding struts **40** align, the struts **40** extend through and past the openings **58**.

(44) FIG. **4b** shows one position of the selector plate **50** relative to the housing **44** of the one-way clutch **32**. In this position, the struts **40** extend past the selector plate **50**, engage, and prevent rotation of the annular plate **36** in the counterclockwise direction, arrow **64**. To prevent rotation of the annular plate **36**, the struts **40** engage openings or recesses (not shown) in the annular plate **36**. With the selector plate **50** positioned as shown in FIG. **4b**, the annular plate **36**, located adjacent to the selector plate **50**, can rotate in the clockwise direction, arrow **66**. The annular plate **36** rotates in the clockwise direction, arrow **66**, because the struts **42** pass over the recesses of the annular plate **36**. In this manner, torque can be transferred in one direction, counterclockwise, but not in the opposite direction, clockwise.

(45) The CMD functions such that when the driving member transfers torque to the driven member in one direction, the struts will become locked in the recesses of the driven coupling plate. The driven coupling plate overruns the driving coupling plate when torque transfer is interrupted. As used herein, the term overrun refers to a condition wherein the struts are up, able to engage, with the plate rotating in the not lock direction-similar to a bicycle traveling forward with pedals stationary; the term freewheel or free refers to a condition wherein the struts are down, not able to engage, with the plate **36** rotating in either direction-similar to a disengaged dog clutch; and the term lock refers to a condition wherein the struts are up and holding torque-similar to engaged dog clutch. In an overrun condition, the struts pass over the recesses of the driven plate, resulting in a planar coupling assembly adapted for torque transfer in one direction while permitting freewheeling relative motion in the opposite direction. The strut selector plate **50** is positioned to permit freewheeling motion.

(46) As illustrated in FIGS. **3a** and **3b**, the CMD **26** includes two one-way clutches **32, 34** acting on each side of the annular plate **36**. Each CMD **26, 28** is a four-mode CMD, the modes being 0/0, 0/1, 1/0, and 1/1. A four-mode CMD provides a lock in both directions 1/1, a lock in one direction 0/1 or 1/0, or a lock in neither direction 0/0. For example, the annular plate **36** can be locked on one side, both sides, or neither side. In the 0/0 mode, unlocked or free, both one-way clutches **32, 34** are

freewheeling. The strut selector plates **50** cover the struts **40, 42** preventing engagement with the annular plate **36**. In mode 1/1, the strut selector plates **50** uncover the struts **40, 42** allowing engagement of the struts with the annular plate **36** and locking the annular plate **36** by preventing rotation of the plate in both clockwise and counterclockwise directions relative to the respective housings of the one-way clutches **32, 34**. Modes 0/1 and 1/0 uncover the struts of one one-way clutch **32, 34**, allowing rotation in one direction only.

(47) As used herein, the term “clutch” should be interpreted to include clutches or brakes wherein one of the plates is drivably connected to a torque delivery element of a transmission, and the other plate is drivably connected to another torque delivery element or is anchored and held stationary to a housing. The terms “coupling,” “clutch,” and “brake” may be used interchangeably.

(48) The illustrated vehicle transmission **12** uses one-way clutches and an electric motor. The transmission **12** provides available features such as park-lock, hill-hold, and neutral. The transmission **12** includes a 1.sup.st gear and a 2.sup.nd gear. The terms 1.sup.st and 2.sup.nd gear(s) apply to forward gears having different gear ratios. The 2-speed electric vehicle transmission **12** has two forward speeds, the first associated with the 1.sup.st gear and the second associated with the 2.sup.nd gear.

(49) The transmission **12** uses a 4-node planetary gearset **22, 24**, an electric motor **10**, and two 4-mode CMDs **26, 28** to shift between a 1.sup.st forward gear and a 2.sup.nd forward gear or from a 2.sup.nd forward gear to a 1.sup.st forward gear using a change in motor direction. For example, when the transmission **12** is in the 1.sup.st forward gear, the electric motor **10** rotates counterclockwise for forward vehicle direction. When the transmission **12** is in the 2.sup.nd forward gear, the electric motor **10** rotates clockwise for forward vehicle direction.

(50) Using the CMDs **26, 28** allows for a quick shift by coming off and landing on the one-way clutch. The CMDs **26, 28** can lock in both directions for power-on and regen in both 1st and 2nd gears. The CMDs **26, 28** allow placing the transmission directly into park, neutral, or hill-hold in either 1.sup.st or 2.sup.nd gear without changing to 1.sup.st gear. The CMDs **26, 28** enable reverse direction in both 1.sup.st and 2.sup.nd gear.

(51) The 2-speed transmission **12**, in connection with changing the rotational direction of the electric motor **10**, provides a 2-speed electric vehicle. Changing the rotational direction of the electric motor **10** enables the vehicle to go forward and reverse in both 1.sup.st and 2.sup.nd gear.

(52) The electric motor **10** rotates and provides power, to the input shaft **62**, in both the counterclockwise and clockwise directions. Correspondingly, the input shaft **62** rotates in the counterclockwise and clockwise directions, arrows **64, 66**. As disclosed herein, the direction of electric motor **10** rotation and selected mode of the CMDs **26, 28** results in different outputs at the output shaft **68**.

(53) As illustrated in FIG. 2, the input shaft **62** provides the input to both the first planetary gearset **22** and the second planetary gearset **24** through a first sun gear (S1) **70** and a second sun gear (S2) **72**. The first planetary gearset **22** includes a plurality of first planetary gears, generally indicated at **74**, a first planet carrier (PC1) **76**, and a first ring or annulus gear (A1) **78**. The second planetary gearset **24** includes a plurality of second planetary gears, generally indicated **80**, a second planet carrier (PC2) **82**, and a second ring or annulus gear (A2) **84**. When the electric motor **10** provides power to the input shaft **62**, in either direction shown by the arrows **64, 66**, the respective first and second sun gears (S1, S2) **70, 72** rotate in the same direction. The output shaft **68** connects to and rotates with the first planet carrier (PC1) **76**. The term “gearset” broadly refers to a gear mechanism for transmitting motion, and in one example includes a set of gears forming a group.

(54) FIG. 2A is a lever diagram illustrating a first lever L1 associated with the 1.sup.st gear and a second lever L2 associated with the 2.sup.nd gear. Each lever has four nodes, the input S1 **70** and S2 **72**, the first CMD **26** associated with A1 **78**, the second CMD **28** associated with PC2 **82**, and the output PC1 **76**, A2 **84**. Depending on the input S1, S2, and where the gearset is grounded, the first or second CMD, the lever diagram represents the output at PC1, A2.

(55) Each CMD **26**, **28** has multiple strut positions, for example, up or down/covered. FIG. 5 is a table showing strut position associated with multiple conditions. The nomenclature (/_) refers to rotation direction, clockwise and counterclockwise (CW/CCW, the first _ refers to the clockwise direction, and the second _ refers to the counterclockwise direction. As indicated, a 1 means: Struts up; either lock or overrun, and a 0 means: Strut down/covered; free in either direction. For example, (1/1): Both strut sets up, lock in both CW or CCW rotation; (1/0): Lock CW rotation, or overrun with CCW rotation; (0/1): Lock CCW rotation, or overrun with CW rotation; and (0/0): Both strut sets down/covered, free in both CW or CCW rotation.

(56) In one example, the transmission **12** includes the first CMD **26** connected to the first ring or annulus (A1) **78**. The connection can be splined, wherein the splines of the annular plate **36** engage complementary splines on the first ring or annulus (A1) **78**. The housing **44** of the CMD **26** is fixed to the transmission housing **30** and remains stationary. The second CMD **28**, like the first CMD **26**, also has its housing **44** fixed to the transmission housing **30** and remains stationary. The annular plate **36** of the second CMD **28** is connected to the second planet carrier (PC2) **82**. FIG. 2 shows a bearing **86** located between the output shaft **68**/first planet carrier (PC1) **76** and the first ring or annulus (A1) **78** and a bearing **87** located between the input shaft **62** and the second planet carrier (PC2) **82**. As illustrated, the first ring or annulus (A1) **78** rotates about the output shaft **68**, and the second planet carrier (PC2) **82** rotates about the input shaft **62**.

(57) The output, rotation speed, and direction of the output shaft **68** can be controlled depending upon the selected mode of each CMD **26**, **28**, which controls the rotation of the first ring or annulus (A1) **78** and the second planet carrier (PC2) **82**.

(58) FIGS. 6-12 show one example of a transmission **12** including two simple planetary gearsets, **22**, **24**, and two CMDs **26**, **28** with the power path, either 1.sup.st or 2.sup.nd gear of the transmission **12**, controlled by rotation direction of the electric motor **10** and the selected mode of each CMD **26**, **28**.

(59) FIG. 6 illustrates the transmission in the park condition. The transmission achieves this condition/position by placing both CMDs **26**, **28** in a 1/1 mode or lock-lock position. The Xs over arrows **64**, **66** indicate that both the input shaft **62** and output shaft **68** cannot rotate either counterclockwise or clockwise. Locking the first ring or annulus (A1) **78** and the second planet carrier (PC2) **82** prevents rotation of the remaining components of the first and second planetary gearsets **22**, **24**, thereby preventing any output or input to or from the wheels **14** of the vehicle. Lever diagram, FIG. 6A shows transmission **12** output at the output node **110** when the vehicle is in the park condition.

(60) FIG. 7 illustrates the transmission in a neutral condition. The transmission achieves this condition/position by placing both CMDs **26**, **28** in a 0/0 mode or free-free position (unlock-unlock). Unlocking the first ring or annulus (A1) **78** and the second planet carrier (PC2) **82** provides free rotation of the remaining components of the first and second planetary gearsets **22**, **24**. Specifically, placing the first ring or annulus (A1) **78** in a free or unlocked position allows rotation of the first ring or annulus (A1) **78** as the first planet carrier (PC1) **76** rotates about the first sun (S1) **70**. As the first planet carrier (PC1) **76** correspondingly rotates the second ring or annulus (A1) **84**, the second planet carrier (PC 2), due to its free or unlocked state, rotates about the second sun **72** (S2) wherein no force, torque, or rotation is applied by or to the input shaft **62**. Lever diagram, FIG. 7A shows transmission **12** output at the output node **110** when the vehicle is in the neutral condition.

(61) FIG. 8 illustrates the transmission placed in a 1.sup.st forward gear with power on and regeneration allowed. The transmission achieves this condition by placing the first CMD **26**, adjacent to the output shaft **68**, in a 1/1 mode or lock-lock position to lock and prevent movement of the first ring or annulus (A1) **78**. Rotating the electric motor **10** in a counterclockwise direction rotates the input shaft **62** in the counterclockwise direction, arrow **64**. Wherein rotation of the sun gear (S1) **70**, acting through the respective first planet gears **74**, rotates the first planet carrier (PC1)

76 and the corresponding output shaft **68** in the counterclockwise direction.

(62) Rotation of the first planet carrier (PC1) **76** correspondingly rotates the second ring or annulus (A2) **84**. The second planet carrier (PC2) **82** is free to rotate because the second CMD **28** is placed in a 0/0 mode or free-free position (unlock-unlock). Allowing rotation of the second planet carrier (PC2) **82** compensates for the difference in rotation between the second ring or annulus (A2) **84** and the second sun (S2) **72**.

(63) The dotted line **90** shows the power flow/path from the input shaft **62** to the output shaft **68**. Because the first CMD **26** is in a 1/1 mode or lock-lock position, during a regeneration phase, the vehicle wheels rotate the output shaft and, through the same power flow/path, dotted line **90**, drive the input shaft wherein the electric motor **10** operates as a generator. For example, the first CMD **26** locks both ways, from the electric motor **10** to the wheels **14** and the wheels **14** to the electric motor **10**.

(64) The lever diagram of FIG. **8A** shows transmission **12** output at the output node **110** when the vehicle is in a 1st forward gear with power on and regeneration allowed. The lever diagram shows the input moving the lever to the right, based on the input of the first sun (S1) with the lever pivoting about the first CMD **26**/first ring or annulus (A1) and the output generated at the first planet carrier (PC1) **76**. In the 1.sup.st forward gear, the first CMD **26** holds/fixes the first ring or annulus (A1) **78** stationary while the input shaft **62** rotates the first sun (S1) **70** in a counterclockwise direction wherein the output is at the output node **110**.

(65) FIG. **9** illustrates the transmission placed in 1.sup.st reverse gear with power on and regeneration allowed. The transmission achieves this condition by placing the first CMD **26**, shown adjacent to the output shaft **68**, in a 1/1 mode or lock-lock position to lock and prevent movement of the first ring or annulus (A1) **78**. Accordingly, when the input shaft rotates the clockwise direction, arrow **66**, rotation of the first sun gear (S1) **70**, acting through the respective first planet gears **74**, rotates the first planet carrier (PC1) **76** and output shaft **68** in a clockwise direction, arrow **66**, and the vehicle moves in reverse, opposite to the forward motion of FIG. **8**.

(66) The lever diagram of FIG. **9A** shows transmission **12** output at the output node **110** when the vehicle is in 1.sup.st reverse gear with power on and regeneration allowed. The lever diagram shows the input moving the lever to the left, based on the input of the first sun (S1) with the lever pivoting about the first CMD **26**/first ring or annulus (A1) and the output generated at the first planet carrier (PC1) **76**. In the 1.sup.st reverse gear, the first CMD **26** holds/fixes the first ring or annulus (A1) **78** stationary while the input shaft **62** rotates the first sun (S1) **70** in a clockwise direction wherein the output is at the output node **110**.

(67) FIG. **8** shows the transmission **12** in the 1.sup.st forward gear, with the input shaft **62** rotated in the counterclockwise direction, arrow **64**. The first CMD **26** is in a 1/1 mode or locked-locked position. This mode holds the first ring or annulus (A1) stationary. The first sun (S1) **70** rotates in the counterclockwise direction, arrow **64**. In this configuration, the first planet carrier (PC1) rotates in the counterclockwise direction, arrow **64**, and rotates the output shaft **68**. The second CMD **28** is in a 0/0 mode, free-free or unlocked-unlocked position.

(68) The 1st gear (FIG. **8**) and the 2nd gear (FIG. **12**) of the transmission **12** can be forward gears; they move the vehicle forward when engaged. The transmission **12** is a 2-speed transmission that shifts between the 1.sup.st and 2.sup.nd gears when both operate as forward gears. For example, when operating in 1.sup.st gear in a forward direction, the transmission **12** may shift to 2.sup.nd gear, which continues the motion of and propels the vehicle forward. Accordingly, the transmission **12** is 2-speed transmission shiftable between a 1.sup.st forward gear and a 2.sup.nd forward gear.

(69) FIG. **12** shows the transmission **12** operating in the 2.sup.nd forward gear. The power flow/path from the input shaft **62** to the output shaft **68**, follows the dotted line **94**. In addition to the different power flow/path from the input shaft **62** to the output shaft **68**, when operating in the 2.sup.nd forward gear, the electric motor **10** rotates in a clockwise direction and correspondingly rotates the input shaft **62** in the clockwise direction, arrow **66**. This is opposite to the

counterclockwise rotation of the electric motor **10** when operating in the 1.sup.st forward gear. While the electric motor **10** rotation is disclosed as counterclockwise for the 1.sup.st forward gear and clockwise for the 2.sup.nd forward gear, this is one example. The respective rotation directions can be switched. The disclosure should not be limited but includes operating one forward gear with or in one motor direction and operating the second, or other forward gear, with or in the other or opposite motor direction.

(70) As set forth more fully herein, shifting the gears between 1.sup.st gear to 2.sup.nd gear or 2.sup.nd gear to 1.sup.st gear occurs when the electric motor **10** changes direction. As disclosed, the transmission **12** operates in the 1.sup.st forward gear when the electric motor **10** rotates in a counterclockwise direction, arrow **64** and in the 2.sup.nd forward gear when the electric motor **10** rotates in a clockwise direction, arrow **66**. The 2-speed transmission shifts between the 1.sup.st forward gear and the 2.sup.nd forward gear when the motor rotation direction changes—from clockwise to counterclockwise or clockwise to counterclockwise.

(71) FIG. **12** shows the transmission **12** shifted from the 1.sup.st forward gear to the 2.sup.nd forward gear and operating in the 2.sup.nd forward gear with the electric motor **10** rotating clockwise. As shown, the input shaft **62** rotates in the clockwise direction, arrow **66**. The second CMD **28** is in a 1/1 mode or locked-locked. This mode holds the second planet carrier (PC2) stationary. The second sun (S2) **72** rotates with the input shaft **62** in the clockwise direction, arrow **66**. When the second CMD **28** is in a 1/1 mode or locked-locked, the second ring or annulus (A2) **84** rotates in a counterclockwise direction, opposite the rotation direction of the second sun (S2) **72**. The second ring or annulus (A2) **84** and the first planet carrier (PC1) **76** are fixed/rotate together and provide the output to the output shaft **68**. Because the first ring or annulus (A1) **78** is unlocked or free, allowed to move, it rotates due to the input of the first sun (S1) **70** and planet gears **74** of the first planet carrier (PC1) **76**. The combined rotation of the first ring or annulus (A1) **78** with the second ring or annulus (A2) **84** and the first planet carrier (PC1) **76** results in an increased output at the output shaft **68**.

(72) The lever diagram of FIG. **12A** shows transmission **12** output at the output node **110** when the vehicle is in a 2nd forward gear with power on and regeneration allowed. The lever diagram shows the input moving the lever, line **104** to the left, based on the clockwise input of the second sun (S2) **72** with the lever pivoting about the second CMD **26**/second planet carrier (PC2) **82** and the output generated at the first planet carrier (PC1) **76** and second ring or annulus (A2) **84**. In the 2.sup.nd forward gear, the second CMD **28** holds/fixes the second planet carrier (PC2) **82** stationary while the input shaft **62** rotates the second sun (S2) **72** in a clockwise direction wherein the output is at the output node **110**.

(73) As shown in FIG. **8**, the transmission **12** operates in the 1.sup.st forward gear along the power flow/path, dotted line **90**, when rotating the electric motor **10** counterclockwise. FIG. **12** shows that the transmission **12** operates in the 2.sup.nd forward gear along the power flow/path, dotted line **94**, when rotating the electric motor **10** clockwise. In 2.sup.nd gear, the vehicle can be driven in reverse, a rearward direction of vehicle travel, by changing the rotation direction of the electric motor **10** to counterclockwise. With the first and second CMDs **26**, **28** placed in the modes shown in FIG. **12**, changing the rotation direction of the motor will change the vehicle direction, either forward or backward. When the transmission **12** is in the 2.sup.nd forward or reverse gear, the second CMD **28** is placed in a 1/1 mode lock-lock condition, and the first CMD **26** is in a 0/0 mode, free-free unlock-unlock condition.

(74) FIGS. **10-11** illustrate an example of the shifting process; how the 2-speed transmission shifts from the 1.sup.st forward first gear to the 2.sup.nd forward gear or from the 2.sup.nd forward gear to the 1.sup.st forward gear based on the rotational direction of the electric motor **10**. As shown in the exemplary embodiment, shifting either from the 1.sup.st forward gear to the 2.sup.nd forward gear or the 2.sup.nd forward gear to the 1.sup.st forward gear is completed with 1 step—changing the motor direction from counterclockwise to clockwise while landing on a one-way clutch in a

power on vehicle forward direction. This can be completed with a layshaft or planetary transmission and with either static or dynamic selectable clutches.

(75) FIGS. **10** and **11** illustrate the modes for the first and second CMDs **26**, **28** during shifting from the 1.sup.st forward gear to the 2.sup.nd forward gear by changing the rotational direction of the electric motor **10**, either from counterclockwise to clockwise or from clockwise to counterclockwise. When shifting, the power flow/path through the transmission **12** changes from the 1.sup.st forward gear power flow/path, dotted line **90**, shown in FIG. **8**, to the 2.sup.nd gear power flow/path, dotted line **94**, shown in FIG. **12**. When the motor changes directions for the shift, the shift comes off a one-way clutch and lands on a different one-way clutch in the direction of power on. In a powershift design, the vehicle has forward torque during the upshift (no torque interruption). In a non-powershift design, the motor typically needs to slow down to synchronize to the output speed when upshifting, engage a clutch, and then resume power-on torque to accelerate the vehicle.

(76) FIG. **10** shows the transmission **12** providing an output in the 1.sup.st forward gear, along the power flow/path shown by dotted line **90**, with the electric motor **10** still applying rotational input in the counterclockwise direction, arrow **64**. In the 1.sup.st forward gear, the first CMD **26** is in a 1/1 mode or lock-lock position. One direction holds torque for forward, power-on vehicle direction, and the other direction holds torque for regeneration and reverse vehicle direction. Before a 1.sup.st to 2.sup.nd gear shift occurs, the first CMD **26**, in a 1/1 mode, goes to a 1/0 mode (one-way clutch mode), and the second CMD **28** goes from a 0/0 mode (free-free position) to a 1/0 mode (one-way clutch mode). In this condition, the vehicle can still go forward, propelled by the electric motor **10**.

(77) When the electric motor **10** transitions from a positive rpm (for example, a counterclockwise direction) to a negative rpm (for example, a clockwise direction), both one-way clutches go into an overrun mode, as shown in the lever diagram of FIG. **13A**, wherein the lever pivots about the output (PC1, A2). As used herein, positive and negative relate to a change in the rotational direction or output of the electric motor **10**. FIG. **13A** shows counterclockwise rotational input as positive and rotational clockwise input as negative. An overrun mode relates to a strut set in an up position, with the plate rotating in the not lock direction, where rotation in the opposite direction is associated with a torque transfer. In an overrun mode or condition, when a driving member transfers torque to a driven member in one direction, the struts will become locked in the recesses of the driven coupling plate. If the driven coupling plate overruns the driving coupling plate, then torque transfer is interrupted, and the struts will pass over the recesses of the driven plate. In this way, a one-way clutch assembly is adapted for torque transfer in one direction while permitting overrun relative motion of the coupling plates in the opposite direction.

(78) FIG. **10** illustrates the first CMD **26** placed in a 1/0 mode, preventing rotation of the first ring or annulus (A1) **78** in a clockwise direction, allowing torque transfer in the counterclockwise direction, arrow **64**, at the output shaft **68**, and permitting freewheeling relative motion of the annular plate **36** of the first CMD **26** and first ring or annulus (A1) **78** in the counterclockwise direction, arrow **64**. As shown, the first ring or annulus (A1) **78** is placed in an overrun condition wherein the first ring or annulus (A1) **78** may rotate faster than the output shaft **68** in the counterclockwise direction. Placing the first CMD **26** in a 1/0 mode locks the rotation of the first ring or annulus (A1) in the clockwise direction while allowing rotation of the first ring or annulus (A1) in the counterclockwise direction, the same direction as the output shaft **68**. FIG. **10** shows power on in first gear, wherein the first CMD **26** holds the first ring or annulus (A1) **78** stationary, against clockwise movement/rotation, enabling power transfer along the power flow/path, shown by the dotted line **90**.

(79) FIG. **10** illustrates the second CMD **28** placed in a 1/0 mode preventing rotation of the second planet carrier (PC2) **82** in a clockwise direction, allowing torque transfer in the counterclockwise direction, arrow **64** at the output shaft **68**, and permitting freewheeling relative motion of the annular plate **36** of the second CMD **28** and the second planet carrier (PC2) **82** in the

counterclockwise direction, arrow **64**. As shown, the second planet carrier (PC2) **82** is placed in an overrun condition wherein the second planet carrier (PC2) **82** may rotate faster than the output shaft **68** in the counterclockwise direction, arrow **64**. Placing the second CMD **28** in a 1/0 mode allows rotation of the second planet carrier (PC2) **82** in the counterclockwise direction, arrow **64**, the same direction as the output shaft **68**, while locking/preventing the rotation of the second planet carrier (PC2) **82** in the clockwise direction, arrow **66**. In this condition, placing the second CMD in a 1/0 mode prepares the transmission **12** to shift from the 1.sup.st forward gear to the 2.sup.nd forward gear. In this condition, the vehicle may still be propelled forward by the electric motor **10**. (80) FIGS. **10** and **11** illustrate respective overrun and lock conditions positions of the first and second CMDs **26**, **28** when the electric motor **10** and correspondingly the input shaft **62** transitions from positive, counterclockwise rotation, arrow **64**, FIG. **10**, to negative, clockwise rotation, arrow **66**, FIG. **11**.

(81) The lever diagram of FIG. **13A** shows the first sun (S1) and second sun (S2) input, rotational direction, and speed along the X-axis **100**. Counterclockwise rotation is on the positive/right side of the Y-axis or node axis **102**, and clockwise rotation is on the negative/left side of the Y-axis or node axis.

(82) The solid line **104** of FIG. **13A**, also shown in FIG. **6A** illustrates the gearset in 1.sup.st gear, setup for shifting to 2.sup.nd gear, with power on in 1.sup.st available, and the mode of the first and second CMDs **26**, **28**. The solid line **104** extends from the first CMD, the first ring or gear (A1)—first CMD node **106**, through the output point **108** of the first planet carrier/second ring or annulus gear (PC1/A2)—output node **110** to input point **112** of the first sun gear (S1)—input node **114**. The second CMD, the second planet carrier (PC2)—second CMD node **116** has a point **118** on line **104**. Before beginning the shift represented by the lever arm, line **104**, the counterclockwise rotation, rotational speed, at point **107**, the intersection of line **104** and y-axis **102** is zero rpm. While the first CMD **26** is in an overrun mode, it is not actually overrunning—as represented by the term “Overrun” associated with the first CMD **26** in FIG. **10**. The counterclockwise rotation, rotational speed, at point **118**, associated with the second CMD **28**, node **116**, is greater than zero rpm resulting in an overrun condition, shown by the term “Overrun,” associated with the second CMD **28**, in FIG. **10**.

(83) The solid line **136** of FIG. **13A**, also shown in FIG. **11**, illustrates the gearset in 2.sup.nd gear, power on in 2.sup.nd gear available, shifted from 1.sup.st gear to 2.sup.nd gear, and the mode of the first and second CMDs **26**, **28**. The solid line **136** extends from the input point **134** of the second sun gear (S2)—input node **114** through the second CMD, the second planet carrier (PC2)—second CMD node **116** having a point **138** on line **136**, the intersection of line **136** and y-axis **102**. The solid line **136** extends through the output point **108** of the first planet carrier/second ring or annulus gear (PC1/A2)—output node **110** to point **140** of the first CMD, the first ring or gear (A1)—first CMD node **106**. In the operating condition represented by the lever arm/line **136**, just after the shift from 1.sup.st gear to 2.sup.nd gear, the counterclockwise rotation, rotational speed, at point **138** is zero rpm, while the second CMD **28** is in an overrun mode, it is not actually overrunning—represented by the term “Overrun” in FIG. **11** associated with the second CMD **28**. The counterclockwise rotation, rotational speed, at output point **140** is greater than zero rpm resulting in an overrun condition, shown by the term “Overrun,” associated with the first CMD **26** in FIG. **11**.

(84) Because the shift from 1st gear to 2nd gear or 2nd gear to 1st gear takes place in under a second and may occur in less than 500 ms, the output—rotational speed of the first planet carrier/second ring or annulus gear combination (PC1/A2) **76**, **84** during the shift operation/procedure remains substantially constant. Typically, the shift operation/procedure takes very little time on the order of 250 ms, and the rotational speed at the output point **108** changes insignificantly. However, because the first and second CMDs **26**, **28** are placed in an overrun condition, they can accommodate changes (reductions or increases) in rotational speed at the output

point **108** during the shift operation/procedure.

(85) FIGS. **13A**, **13C**, and **13D** show transitional lever diagrams/lines as the rotation speed and direction of the electric motor **10** change. Because the output point **108** remains relatively constant during the shift operation/procedure, the lever arm of FIG. **13A** rotates about the output point **108**. The lever arm transitions from the 1.sup.st gear condition, line **104**, FIGS. **10**, **13A** to the 2.sup.nd gear condition, line **136**, FIGS. **11**, **13A** as the rotational speed and direction of the electric motor **10** and the corresponding input shaft **62** changes. As the input shaft **62** rotational speed reduces in the counterclockwise direction, the input point **112** moves along the X-axis **100**, and the lever arm pivots about the output point **108**. When the lever arm reaches the first dashed line **120**, the counterclockwise rotation, rotational speed, at the point **122** of the second CMD, the second planet carrier (PC2)—second CMD node **116**, while less than that at the point **118**, still exceeds 0 rpm, resulting in a continuing overrun condition at the second CMD **28**. The lever arm shown by the first dashed line **120** shows the output at point **124** of the first CMD **26**, first ring or gear (A1)—first CMD node **106**, now exceeds zero rpm, resulting in an overrun condition at the first CMD **26**.

(86) When the motor input reaches the rotational input point **128**, the lever arm pivots about the output point **108** to the position shown by the second dashed line **126**. The rotational input point **128** is near the transition point from counterclockwise motor rotation to clockwise motor rotation—the intersection of the x-axis **100** and y-axis **102**, input node **114**, wherein the input changes from counterclockwise to clockwise and from the first sun (S1) to the second sun (S2). Lever arm/line **126** shows the counterclockwise rotation and rotational speed at point **131** of the second CMD, the second planet carrier (PC2)—second CMD node **116**, still exceeds 0 rpm, and point **130** of the first CMD **26**, the first ring or gear (A1)—first CMD node, also exceeds zero rpm, resulting in an overrun condition at the first CMD **26** and the second CMD **28**.

(87) The dashed line **132** of FIG. **13A** shows further progression of the lever arm toward the 2.sup.nd gear input point **134** associated with a lever arm shown by solid line **136**. As the input moves from the first sun (S1)—input point **112** to the second sun (S2)—input point **134**, arrow **142**, the lines **120**, **126**, and **132** pivot about the output point **108**, showing the overrun speed of the first CMD **26**, the first ring or gear (A1)—first CMD node **106**, increasing from 0 rpm, arrow **144** and the overrun speed of the second CMD, the second planet carrier (PC2)—second CMD node **116** decreasing towards 0 rpm. As illustrated, during a shift between the 1.sup.st and 2.sup.nd gear, when the motor input moves between the first and second sun (S1, S2) input points **112**, **134**, both the first and second CMDs **26**, **28** are in an overrun condition.

(88) At input point **134**, the second CMD **28** engages at the second CMD node **116**, whereby increased clockwise rotation at the input shaft **62** generates an increased output at output point **108**. The one-way clutch of the second CMD **28** will decrease in overrun speed starting at the higher rpm until it captures the lever arm represented by solid line **136**, FIG. **13A** and FIG. **13E**-lever diagram, as the lever arm attempts to pass through the second CMD node **116**. As the motor begins negative, clockwise, rpm, the overrunning clutch of the second CMD **28** captures the spinning second planet carrier (PC2) **82**, preventing clockwise rotation by landing on a one-way clutch in the power on vehicle forward direction.

(89) Once holding torque on the 1/0 one-way clutch—second CMD **28**, the second CMD **28** opens a second selector plate changing the one-way clutch (1/0) into a lock-lock clutch (1/1). With the transmission **12** now in the 2.sup.nd forward gear, the second CMD **28** holds torque in both directions, one direction for forward vehicle power-on direction and the other direction for regeneration or reverse vehicle direction. The struts of the first CMD **26** are placed in a 0/0 configuration or covered/free in both the clockwise or counterclockwise rotation.

(90) The transmission **12** may achieve a park condition by placing both CMDs **26**, **28** into 1/1 mode (lock-lock). The transmission **12** can be placed directly in the park condition, regardless of whether the transmission **12** is in the 1.sup.st forward or the 2.sup.nd forward gear. In addition, the transmission **12** can move directly from the park to either the 1.sup.st forward or the 2.sup.nd

forward gear. The disclosed transmission **12** may also move directly to reverse from the 1.sup.st forward or the 2.sup.nd forward gear. With the disclosed 2-speed transmission it is unnecessary to shift from the 2nd forward gear to the 1st forward gear before placing the 2-speed transmission in either park or reverse. The first and second CMDs **26**, **28** also provide a hill-hold function in either the 1.sup.st forward gear or the 2.sup.nd forward gear for both up-hill and down-hill hill-hold since each strut set is independently controlled. As shown in FIG. **3**, neutral is achieved by putting each CMD **26**, **28** into a 0/0 mode or free-free position.

(91) As disclosed, when the transmission is ready to shift, the electric motor quickly goes from a positive rpm to a negative rpm. FIGS. **6-12** illustrate the positive rpm associated with the counterclockwise direction and the negative rpm associated with the clockwise direction. As explained, these directions can be reversed or opposite. FIGS. **13A-13E** illustrate lever diagrams of the transmission **12** as it progresses through the shift procedure, from the 1.sup.st forward gear to the 2.sup.nd forward gear. FIG. **13B** is a lever diagram of the 2-speed transmission **13** in the 1.sup.st forward gear. FIG. **13C** illustrates a lever diagram of the 2-speed transmission shifting from the 1.sup.st forward gear to the 2.sup.nd forward gear, with the lever pivoting about the output node (PC1, A2), with the first CMD **26** in a 1/0 mode at node A1 and the second CMD **28** in a 1/0 mode at node PC2 as the rpm is reduced. FIG. **13D** illustrates a lever diagram wherein the first CMD **26** goes to a 0/0 mode. The second CMD **28**, node PC2, captures the lever just after it passes through zero (0) rpm. FIG. **13E** illustrates a lever diagram wherein, now in the 2.sup.nd forward gear, the second CMD **28** holds torque in both the forward vehicle power-on direction and the regeneration or reverse vehicle direction.

(92) One example of the vehicle drive system **8** includes using an electric motor **10** having a motor rotor inertia: 0.03 kgm.sup.2, a maximum e-motor speed: 10,000 rpm; a max vehicle speed: 100 mph, a ratio spread between 1.sup.st and 2.sup.nd gear: 2; and a shift speed between 1.sup.st and 2.sup.nd gear: 40 mph. The electric motor **10** is slowed from a 40 mph 1.sup.st gear e-motor speed: 4,000 rpm CCW, with a target time to slow down from 4,000 rpm to 0 rpm of 132 ms. The electric motor **10** then changes rotation direction, from counterclockwise (CCW) to clockwise (CW), and speeds up to a 40 mph 2nd gear e-motor speed of 2,000 rpm CW. The target time to speed up to 2,000 rpm from 0 rpm of 33 ms resulting in a Total shift time of 165 ms.

(93) FIG. **14** illustrates an alternative example of the transmission **12** using a Ravigneaux gearset with associated CMDs to achieve a multi-speed transmission that shifts based on a change in motor rotation direction. As illustrated, the input shaft **250** provides the input to the Ravigneaux gearset, generally indicated at **252**, through a second sun gear (S2) **254**. The Ravigneaux gearset **252** includes a plurality of planetary gears, generally indicated at **256**, a planet carrier (PC) **258**, and a ring or annulus gear (A1) **260**. When the electric motor **10** provides power to the input shaft **250**, in either direction, as shown by the arrows **262**, **264**, the second sun gear (S2) **254** rotates in the same direction. The output shaft **268** connects to and rotates with the planet carrier (PC) **258**. The planet carrier (PC) **258** is rotatably mounted about the output shaft **268** on a bearing **273**. The first sun gear (S1) **270** is rotatably mounted about the input shaft **250** on a bearing **272**. As illustrated in FIG. **14**, the controllable mechanical diodes, CMDs **26**, **28** located on each side of the Ravigneaux gearset **252** operate to selectively couple portions of the Ravigneaux gearset **252** to a transmission housing, generally indicated at **30**.

(94) FIG. **14A** is a lever diagram illustrating a first lever L1 associated with the 1.sup.st forward gear and a second lever L2 associated with the 2.sup.nd forward gear, wherein the horizontal force and velocity relationship of the levers are identical to the torque and rotational velocity relationship of the gearset. Each lever has four nodes, the input at the second sun gear (S2) **254**, the first CMD **26** associated with the first sun gear (S1), the second CMD **28** associated with the ring or annulus gear (A) **260**, and the output at the planet carrier (PC) **258**. Depending on the input (S2) and where the gearset is grounded, the first or second CMD **26**, **28**, the lever diagram represents the output at the planet carrier (PC) **258**.

(95) FIGS. 15A-15F are further lever diagrams illustrating the respective modes of each CMD 26, 28 at the various nodes. Lever diagram FIG. 15A shows the transmission output at the output node-planet carrier (PC) 258 when the vehicle is in the park condition. The transmission 12 achieves this condition/position by placing both CMDs 26, 28 in a 1/1 mode or lock-lock position. Locking the first sun gear (S1) 270 and the planet carrier (PC) 258 prevents rotation of the remaining components of the Ravigneaux gearset 252, thereby preventing any output or input to or from the wheels 14 of the vehicle.

(96) The lever diagram of FIG. 15B shows the transmission output at the output node-planet carrier (PC) 258 when the transmission 12 is in the neutral condition. The transmission 12 achieves this condition/position by placing both CMDs 26, 28 in a 0/0 mode or free-free position (unlock-unlock). Unlocking the ring or annulus (A) 260 and the planet carrier (PC) 258 provides free rotation of the remaining components of the Ravigneaux gearset 252, wherein no force, torque, or rotation is applied at the second sun (S2) 254 by or to the input shaft 250.

(97) Lever diagram 15C shows transmission 12 output at the output node—planet carrier (PC) 258 when the vehicle is in a 1.sup.st forward gear with power on and regeneration allowed. The lever diagram shows moving the lever to the right, based on the input of the second sun (S2), with the lever pivoting about the first CMD/first sun (S1) 270 and the output generated at the output node—planet carrier (PC) 258. In the 1.sup.st forward gear, the first CMD 26 holds/fixes the first sun (S1) stationary while the input shaft 250 rotates the second sun (S2) 254 in a counterclockwise direction wherein the output is at the output node—planet carrier (PC) 258.

(98) Lever diagram 15D shows transmission 12 output at the output shaft 268 when the vehicle is in 1.sup.st reverse gear with power on and regeneration allowed. The lever diagram shows the input moving the lever to the left, based on the input of the second sun (S2) 254, with the lever pivoting about the first CMD 26/first sun (S1) 270 with the output generated at output node—the planet carrier (PC) 258. In the 1.sup.st reverse gear, the first CMD 26 holds/fixes the first sun (S1) 270 stationary while the input shaft 250 rotates the second sun (S2) 254 in a clockwise direction with the output generated at the output node—the planet carrier (PC) 258.

(99) Lever diagram FIG. 15E shows transmission 12 output at the output node—the planet carrier (PC) 258 when the vehicle is in 2.sup.nd forward gear with power on and regeneration allowed. The lever diagram shows the input moving the lever to the left, based on the clockwise input of the second sun (S2) 254 with the lever pivoting about the second CMD 28/ring or annulus (A) 260 and the output generated at the output node—the planet carrier (PC) 258. In the 2.sup.nd forward gear, the second CMD 28 is in a 1/1 mode, or lock-lock position, and the first CMD 26 is in a 0/0 mode or free-free position (unlock-unlock) wherein the first sun (S1) 270 rotates freely in a counterclockwise direction with the output generated at the output node—the planet carrier (PC) 258.

(100) Lever diagram FIG. 15F shows the 2-speed transmission shifting from the 1.sup.st forward gear to the 2.sup.nd forward gear, wherein the lever pivots about the output node—the planet carrier (PC) 258, with the second CMD 28 in a 1/0 mode at ring or annulus (A) 260 and the first CMD 26 in a 1/0 mode at the first sun (S1) 270. As the rpm is reduced, the input at the second sun (S1) moves to the left, from that shown in FIG. 15C, and reaches zero input, zero (0) rpm in the counterclockwise direction, wherein the first CMD 26 goes to a 0/0 mode and the second CMD 28, at the ring or annulus (A) 260, catching the lever after it passes through zero (0) rpm, see FIG. 15E. As shown in FIG. 15E with the transmission 12 in the 2.sup.nd forward gear, the second CMD 28 in a 1/1 mode holds torque in both the forward vehicle power-on direction and the regeneration or reverse vehicle direction.

(101) FIGS. 16-18 show alternative embodiments of the transmission 12 using different gear arrangements or configurations. Each embodiment shifts from a 1.sup.st forward gear to a 2.sup.nd forward gear based on a change in motor rotation.

(102) FIG. 16 illustrates an alternative example of the transmission 12 using a layshaft

configuration. The layshaft configuration includes a first or input shaft **200** and a second or output shaft **202**. The first shaft **200**, includes input gears **204**, **206**. The second or output shaft **202** includes output gears **208**, **210**. An intermediate gear **212** is positioned between input gear **204** and output gear **208**. A dynamically controllable clutch (DCC), seen schematically at **214**, one wherein both races of the clutch can rotate, operates to selectively couple one of the input gears **204**, **206** to the first or input shaft **200** and correspondingly drive the second or output shaft **202**. The dynamically controllable clutch **214** is a multi-mode clutch; in one example, a four-mode clutch. (103) FIG. **16A** illustrates another example of the transmission **12** using a layshaft configuration, similar to that shown in FIG. **16**. As shown, adding a second 4-mode dynamically controllable clutch (DCC) **215** between gears **208** and **210** enables selective coupling or uncoupling from output shaft **202**. Adding the second 4-mode dynamically controllable clutch (DCC) **215** removes the unused power-flow direction from the geartrain for improved efficiency.

(104) FIG. **17** shows another alternative example of the transmission **12** using a simple or single planetary gearset **220**, a first clutch assembly **222**, and a second clutch assembly **224**. The first clutch assembly **222** is a controllable mechanical diode (CMD) that selectively couples a planet carrier (PC) **226** to a housing **228**. The second clutch assembly **224** is a dynamically controllable clutch (DCC) having an inner race (DCC-IR) **238** and an outer race (DCC-OR) **240**. The inner race (DCC-IR) **238** connects to the planet carrier (PC) **226**, and the outer race (DCC-OR) **240** connects to the input shaft **230**. The second clutch assembly **224**, or dynamically controllable clutch (DCC), selectively couples the input shaft **230** to the planet carrier (PC) **226**. The output shaft **232** is connected to the ring or annulus (A) **234**.

(105) The lever diagram of FIG. **17A** shows transmission **12** output at the output node—the ring or annulus (A) **234** when the transmission **12** is in a 1.sup.st forward gear and a 2.sup.nd forward gear. The lever diagram shows the input moving the lever to the left in 1.sup.st forward gear, based on the input of the input shaft **230** and corresponding sun gear(S) **236**, with the lever pivoting about the first clutch assembly or CMD **222** with the output at the output node—the ring or annulus (A) **234**. In the 1.sup.st forward gear, the first clutch assembly or CMD **222** holds/fixes the planet carrier **226** stationary while the input shaft **230** rotates the sun(S) **236** in a clockwise direction wherein the output is at the output node—the ring or annulus (A) **234**. The lever diagram shows the input moving the lever to the right in the 2.sup.nd forward gear based on the input of the input shaft **230** and corresponding sun gear(S) **236**. When in the 1.sup.st forward gear, the output at the output shaft **232** varies, dependent upon the gear ratios, from the input at the input shaft **230**. In the 2.sup.nd forward gear, the second clutch assembly or DCC **224** holds/fixes the input shaft **230** and the planet carrier (PC) **226**, wherein the input shaft **230** and planet carrier **226** rotate together. The ring or annulus (A) **234** rotates with the input shaft **230** and planet carrier **226**. When in the 2.sup.nd forward gear, the output at the output shaft **232** equals the input at the input shaft **230**.

(106) FIG. **18** shows a further alternative example of the transmission **12** using a simple or single planetary gearset **220**, a first clutch assembly **222**, and a second clutch assembly **224**. The example of FIG. **18** differs from that of the example shown in FIG. **17** in that the input and outputs are reversed. The input shaft **230** is connected to the ring or annulus (A) **234**, with the output shaft **232** connected to the sun(S) **236** of the simple or single planetary gearset **220**. Like the previous embodiment, the first clutch assembly **222** is a controllable mechanical diode (CMD) that selectively couples a planet carrier (PC) **226** to a housing **228**, and the second clutch assembly **224** is a dynamically controllable clutch (DCC) having an inner race (DCC-IR) **238** and an outer race (DCC-OR) **240**. The inner race (DCC-IR) **238** connects to the planet carrier (PC) **226**, and the outer race (DCC-OR) **240** connects to the output shaft **232**. The second clutch assembly **224**, or dynamically controllable clutch (DCC), selectively couples the output shaft **232** to the planet carrier (PC) **226**.

(107) The lever diagram of FIG. **18A** shows transmission **12** output at the output shaft **232** when the transmission **12** is in a 1.sup.st forward gear and a 2.sup.nd forward gear. The lever diagram

shows the input, rotating counterclockwise, moving the lever to the right in the 1.sup.st forward gear based on the input of the input shaft **230** and corresponding ring or annulus (A) **234**. In the 1.sup.st forward gear, the second clutch assembly or DCC **224** holds/fixes the output shaft **232** and the planet carrier **226**, wherein the output shaft **232** and planet carrier **226** rotate together. The ring or annulus (A) **234** rotates with the input shaft **230** and planet carrier **226**. As shown, in the 1.sup.st forward gear the output at the output node, the output at the output shaft **232**, equals the input at the input shaft **230**/ring or annulus (A) **234**. The lever diagram shows the input moving the lever to the right in 1.sup.st forward gear based on the input of the input shaft **230** and corresponding ring or annulus (A) **234**. In the 2.sup.nd forward gear, based on the input of the input shaft **230** and corresponding ring or annulus (A) **234**, the lever pivoting about the CMD **222** with the output at the output node—the sun gear(S) **236** and corresponding output shaft **232**. In the 2.sup.nd forward gear, the CMD **222** holds/fixes the planet gear carrier **226** stationary while the input shaft **230** rotates the ring or annulus (A) **234** in a clockwise direction wherein the output is at the sun(S). The lever diagram shows the input, rotating clockwise, moves the lever to the left in the 2.sup.nd forward gear. The lever diagram shows that in the 2.sup.nd forward gear the output at the output node—sun(S) **236** varies, dependent upon gear ratios, from the input at the input shaft **230**.

(108) The 2-speed transmission of the exemplary embodiment uses two controllable one-way clutches and no friction clutches to achieve a 2-speed transmission for an electric vehicle.

Changing the rotational direction of the electric motor **10** enables, during the shift, the power transfer to come off a one-way clutch and land on a different one-way clutch in the direction of power on. This allows the transmission to be easily controlled, have low drag losses, and have small packaging.

(109) The foregoing applies to 3, 4, and 5-speed transmissions. A 2-speed transmission uses a 4-node planetary gearset. When using this design for a 3, 4, or 5-speed transmission, a 3-speed would use a 5-node planetary, a 4-speed would use a 6-node planetary, and a 5-speed would use a 7-node planetary, etc. Another example of a 3-speed transmission adds a dynamically controllable clutch (DCC) to a 4-node planetary gearset.

(110) A further example of a 4-node planetary gearset includes changing the input at the outer nodes and the output at the inner nodes. For example, using the lever diagram of FIG. 2A, instead of the input, input node **114**, being the first and second sun gears (S1, S2) **70, 72**, the input node **114** moves to the first ring or gear (A1) **78**. Correspondingly, the first CMD node **106**, the first CMD **26**, moves to the first and second sun gears (S1, S2) **70, 72**. The output node **110**, moves from the first planet carrier/second ring or annulus gear (PC1, A2) **76, 84** to the second planet carrier (PC2) **82**. Correspondingly, the second CMD node **116** moves to the first planet carrier/second ring or annulus gear (PC1, A2) **76, 84**. The foregoing input and output configuration also works with a Ravigneaux gearset.

(111) The exemplary 2-speed transmission can be used as a high/low transmission since park, hill-hold, and reverse can be achieved directly from either the 1.sup.st or the 2.sup.nd gear. The 2-speed transmission of the disclosed embodiments can have full function while staying in the 1.sup.st or 2.sup.nd gear. High torque density and similar design can achieve 3-speeds if desired.

(112) The description of the invention is merely exemplary in nature. Thus, variations that do not depart from the gist of the invention are intended to be within the scope of the invention. Such variations are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. A vehicle drive system comprising: an electric motor operable in a first direction and a second direction; a first shaft; a second shaft; a first gearset connecting the first shaft to the second shaft, the first gearset includes an input gear associated with the first shaft and an output gear associated with the second shaft, with an intermediate gear positioned between the input gear and the output

gear; a second gearset connecting the first shaft to the second shaft, the second gearset includes an input gear associated with the first shaft and an output gear associated with the second shaft; and a controllable clutch selectively coupling one of the input gear of the first gearset and the input gear of the second gearset to the first shaft wherein operating the electric motor in the first direction and selectively coupling the input gear of the first gearset to the first shaft operates the second shaft at a first forward speed and operating the electric motor in the second direction and selectively coupling the input gear of the second gearset to the first shaft operates the second shaft at a second forward speed.

2. The vehicle drive system of claim 1 wherein: the controllable clutch is a dynamically controllable clutch wherein the dynamically controllable clutch connects the first shaft with one of the input gear of the first gearset and the input gear of the second gearset wherein the first shaft and respective input gear rotate together.

3. The vehicle drive system of claim 1 wherein: the controllable clutch is a multi-mode clutch.

4. The vehicle drive system of claim 3 wherein: the multi-mode clutch is a four-mode clutch.

5. The vehicle drive system of claim 1 wherein: the controllable clutch includes a plurality of selectable clutches.

6. The vehicle drive system of claim 1 wherein: the controllable clutch includes a first multi-mode clutch connected to the first shaft and the input gear of the first gearset and a second multi-mode clutch connected to the second shaft and the input gear of the second gearset.

7. The vehicle drive system of claim 1 including: a second controllable clutch, the second controllable clutch selectively couples one of the output gear of the first gearset and the output gear of the second gearset to the second shaft.

8. A vehicle drive system comprising: a motor operable in a first direction and a second direction; a transmission having an input and an output, the input connected to the motor; a first gearset including an input gear associated with the input, an output gear associated with the output, and an intermediate gear positioned between the input gear and the output gear; and a second gearset including an input gear associated with the input and an output gear associated with the output; a first controllable multi-mode clutch selectively coupling the input gear of the first gearset and the input; and the first controllable multi-mode clutch selectively coupling the input gear of the second gearset and the input wherein the transmission has a first forward speed and a second forward speed, the transmission operating in the first forward speed when the motor operates in the first direction of rotation and operates in the second forward speed when the motor operates in the second direction of rotation.

9. The vehicle drive system of claim 8 including: a second controllable multi-mode clutch selectively coupling the output gear of the first gearset and the output; and the second controllable multi-mode clutch selectively coupling the output gear of the second gearset and the output.

10. The vehicle drive system of claim 9 wherein: the first controllable multi-mode clutch has at least four modes.

11. The vehicle drive system of claim 9 wherein: the second controllable multi-mode clutch has at least four modes.

12. The vehicle drive system of claim 9 wherein: the direction of motor operation determines the selective coupling of the first controllable multi-mode clutch and the second controllable multi-mode clutch.

13. A transmission comprising: an input shaft rotatable in a first direction and a second direction; an output shaft; a first gearset including an input gear associated with the input shaft, an output gear associated with the output shaft, and an intermediate gear positioned between and engaging the input gear and the output gear; a second gearset includes an input gear associated with the input shaft and an output gear associated with the output shaft; a first clutch selectively coupling the input gear of the first gearset to the input shaft; the first clutch selectively coupling the input gear of the second gearset to the input shaft; a second clutch selectively coupling the output gear of the

first gearset to the output shaft; and the second clutch selectively coupling the output gear of second gearset to the output shaft wherein the transmission operates in a first gear when the input shaft rotates in the first direction and operates in a second gear when the input shaft rotates in the second direction.

14. The transmission of claim 13 wherein: the first clutch includes a dynamically controllable clutch positioned between the input gear of the first gearset and the input gear of the second gearset; and the second clutch includes a dynamically controllable clutch positioned between the output gear of the first gearset and the output gear of the second gearset.

15. The transmission of claim 13 wherein: the transmission has a first forward speed and a second forward speed, the transmission operating in the first forward speed and the second forward speed based on the rotation direction of the input shaft and the selective coupling of the first clutch and the second clutch.

16. The transmission of claim 13 wherein: the first clutch includes a dynamically controllable clutch connecting the input shaft and one of the input gear of the first gearset and the input gear of the second gearset wherein the input shaft and respective input gear rotate together; and the second clutch includes a dynamically controllable clutch connecting the output shaft and one of the output gear of the first gearset and the output gear of the second gearset wherein the output shaft and the respective output gear rotate together.

17. The transmission of claim 13 wherein: the first clutch is a multi-mode clutch and the second clutch is a multi-mode clutch.

18. A transmission comprising: an input shaft rotatable in a first direction and a second direction; an output shaft; a first gearset including an input gear associated with the input shaft, an output gear associated with the output shaft, and an intermediate gear positioned between and engaging the input gear and the output gear; a second gearset includes an input gear associated with the input shaft and an output gear associated with the output shaft; a first clutch selectively coupling one of the input gear of the first gearset and the input gear of the second gearset to the input shaft; a second clutch selectively coupling one of the output gear of the first gearset and the output gear of second gearset to the output shaft wherein the transmission operates in a first gear when the input shaft rotates in the first direction and operates in a second gear when the input shaft rotates in the second direction; the first clutch includes a first multi-mode clutch having four modes and a second multi-mode clutch having four modes; and the second clutch includes a first multi-mode clutch having four modes and a second multi-mode clutch having four modes.
